Beyond Average Return: Federated Reinforcement Learning via Adaptive Minimax Optimization

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***Abstract*—Federated reinforcement learning (FRL) aims to learn a shared policy across distributed environments without centralized data collection. Existing methods optimize an average-return objective, effectively optimizing performance in a virtual environment induced by a mixture of local transition dynamics, which provides no per-environment performance guarantee under environment heterogeneity. To address this limitation, we propose a robust FRL framework based on a minimax objective over environments. A key challenge is that the worst-case environment is *policy dependent* and evolves during training, rendering static robust formulations fundamentally inadequate. We reformulate the problem as a saddle-point optimization and develop a primal–dual policy gradient method, termed *adaptive minimax policy optimization* (AMPO), which adaptively reweights environments to track those that limit worst-case performance. We establish sublinear convergence to a first-order stationary point and sublinear regret for the dual variable, providing a principled guarantee that the policy continuously improves on the most challenging environments. Empirical evaluations on heterogeneous benchmark environments confirm that AMPO significantly improves worst-case performance while maintaining competitive average performance.**

***Index Terms*—Federated reinforcement learning, robust optimization, saddle-point methods, environment heterogeneity.**

I. Introduction

Learning a single policy that performs reliably across diverse environments is a central challenge in reinforcement learning (RL), particularly in distributed settings with heterogeneous dynamics. In such scenarios, environment heterogeneity can lead to significant performance variability across agents, making robust policy learning fundamentally difficult.

Federated reinforcement learning (FRL) has emerged as a promising paradigm to address this challenge by combining RL with federated learning (FL) [1]–[3]. In FRL, multiple agents interact with their own local environments and collaboratively learn a shared global policy through periodic aggregation of local updates, without sharing raw trajectories. This framework improves sample efficiency while preserving data privacy, making it particularly attractive for distributed systems.

A fundamental challenge in FRL is *environment heterogeneity*. Unlike standard RL, which optimizes a policy within a single environment, FRL requires learning a shared policy across multiple environments with possibly distinct transition dynamics. This induces a fundamental trade-off: improving performance in one environment may degrade performance in others. Moreover, the state–action visitation distribution depends on the policy itself, making the heterogeneity inherently *policy-dependent*. This distinguishes FRL not only from standard RL, but also from conventional FL, where heterogeneity is fixed and arises solely from differences in data distributions, independent of the learned policy.

Most existing FRL approaches optimize an *average-return* objective, evaluating policies by averaging returns across environments. This formulation is inherited from the empirical risk minimization (ERM) paradigm of FL, where averaging over clients is the standard aggregation strategy. However, it is fundamentally misaligned with deployment: a policy is executed in a *single* environment, not an average of them. More critically, it effectively optimizes performance in a *virtual* environment induced by a mixture of local transition dynamics, which may not correspond to any real environment. As a result, policies trained under this objective can perform well on this virtual environment yet fail catastrophically in individual environments, especially under strong heterogeneity.

These limitations reveal a fundamental misalignment: optimizing average return provides no guarantee on the performance of any individual environment. Since a deployed policy must perform reliably in *each* environment rather than on average, a natural and principled remedy is to directly optimize the *worst-case* return across environments–ensuring that even the most challenging environment achieves satisfactory performance. However, existing FRL methods doe not pursue this goal. To address this, we propose a robust FRL framework that directly optimizes the worst-case return:

$$\max_{\pi} \min_{k\in[K]} J_k(\pi), \tag{1}$$

where $J_k(\pi)$ denotes the discounted return of policy π under environment k. This minimax objective aligns training with the deployment requirement of reliable per-environment performance, explicitly accounting for the most challenging environment at each stage of learning.

A key difficulty is that the worst-case environment in (1) is inherently *policy-dependent*: the index $k^\star = \arg\min_k J_k(\pi)$ may shift as the policy evolves, inducing a non-stationary optimization landscape that invalidates fixed worst-case formulations used in classical robust optimization. To address this, we reformulate the problem as a saddle-point optimization via a primal–dual framework, where the dual variable dynamically

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reweights environments to track the policy-dependent worst-case environment throughout training.

While our formulation shares a minimax structure with classical robust RL, the underlying interpretation is fundamentally different. In robust RL, the worst-case is defined over a fixed uncertainty set of transition dynamics [REFs]. In contrast, our setting considers a finite collection of heterogeneous environments where the worst-case environment is policy-dependent and evolves during training. Our framework thus captures a *dynamic* notion of robustness of performance-limiting environments rather than optimizing against a fixed adversarial scenario.

A. Related Works

Jin *et al.* [4] studied FRL under environment heterogeneity and proposed QAvg, a federated extension of Q-learning based on the standard averaging paradigm. Wang *et al.* [5] developed momentum-based policy gradient methods, including FedSVRPG-M and FedHAPG-M, to improve optimization efficiency via variance reduction and acceleration. Labbi *et al.* [6] established convergence guarantees for federated policy gradient methods under heterogeneous environments, achieving near-optimal stationary convergence with linear speedup. Zhu *et al.* [7] proposed a single-loop federated actor–critic (SFAC) algorithm that jointly updates actor and critic in a unified framework.

Despite differences in algorithmic design and theoretical guarantees, existing FRL methods predominantly optimize an *average-return* objective. As a result, none provide explicit guarantees on per-environment performance, motivating the robust formulation proposed in this work.

B. Contributions

We propose a robust FRL framework that departs from the standard average-return paradigm by adopting a minimax objective in (1). Our key insight is that robustness emerges from *adaptive reweighting* of environments, enabling the learning process to dynamically focus on performance-limiting regimes. Our main contributions are summarized as follows:

- We introduce a minimax formulation (1) for FRL, aligning the training objective with the deployment requirement of reliable per-environment performance and addressing the fundamental limitation of average-return objectives.
- We cast the problem as a saddle-point optimization via a primal–dual framework, where the dual variable dynamically reweights environments to track the policy-dependent worst-case environment. Based on this formulation, we develop an *adaptive minimax policy optimization* algorithm, termed AMPO.
- We establish sublinear convergence to a first-order stationary point of a dynamically reweighted objective and sublinear regret for the dual variable, providing theoretical guarantees that the policy steadily improves on the most challenging environments while the dual weighting reliably tracks performance-limiting regimes throughout training.
- We demonstrate empirically that AMPO improves worst-case performance while maintaining competitive average performance across heterogeneous environments.

II. From Average to Worst-Case: Robust FRL Formulation

We consider a federated reinforcement learning (FRL) setting with K environments (clients), each characterized by a distinct transition kernel P_k. The objective is to learn a single global policy π that generalizes across all environments. The discounted return of policy π under environment P_k is defined as

$$J_k(\pi) := \mathbb{E}_{P_k}\left[\sum_{t=0}^{T-1} \gamma^t r(s_t, a_t)\right], \tag{2}$$

where $\gamma \in (0, 1)$ denotes the discount factor.

Conventional FRL approaches optimize the average return across environments [4]:

$$\frac{1}{K}\sum_{k=1}^{K} J_k(\pi). \tag{3}$$

As discussed in Section I, this objective is misaligned with deployment: it effectively maximizes performance in a *virtual environment* induced by a mixture of local transition dynamics, providing no guarantee of satisfactory performance in any individual environment.

Motivated by this limitation, we introduce a the following minimax formulation that directly optimizes worst-case performance:

$$\max_{\pi} \min_{k\in[K]} J_k(\pi). \tag{4}$$

This formulation enforces robustness by directly optimizing the worst-case return, ensuring reliable performance across all environments. However, the worst-case environment is inherently *policy-dependent*: the minimizing index $k^\star = \arg\min_k J_k(\pi)$ may shift as the policy evolves, rendering static robust formulations inadequate.

To address this, we adopt a primal–dual approach in which the dual variables define a distribution over environments, converting the non-smooth worst-case objective into a tractable weighted objective that evolves with the policy. The policy (primal variable) is optimized with respect to dynamically evolving environment weights, while the dual variables are updated to emphasize environments with lower returns. This closed-loop interaction reveals an *adaptive minimax optimization* structure, in which learning continuously focuses on performance-limiting environments. The formal primal–dual reformulation is developed in the next section.

III. Primal–Dual Reformulation for Robust FRL

We present a primal–dual optimization framework for solving the robust FRL problem in (4), which provides a principled foundation for the adaptive minimax algorithm developed in subsequent sections.

A. Primal-Dual Formulation

To facilitate optimization, we introduce an auxiliary variable v and rewrite (4) in epigraph form:

$$\begin{aligned} \max_{\pi,\, v} \quad & v \\ \text{s.t.} \quad & J_k(\pi) \geq v, \quad \forall k \in [K], \end{aligned} \tag{5}$$

where v represents the worst-case return across environments. Maximizing v therefore corresponds to improving worst-case performance.

Applying Lagrangian relaxation to (5) with multipliers $\lambda_k \geq 0$ yields

$$\mathcal{L}(\pi, v, \lambda) = v + \sum_{k=1}^{K} \lambda_k \big(J_k(\pi) - v\big), \qquad \lambda_k \geq 0. \tag{6}$$

Rearranging terms yields

$$\mathcal{L}(\pi, v, \lambda) = \sum_{k=1}^{K} \lambda_k J_k(\pi) + \Big(1 - \sum_{k=1}^{K} \lambda_k\Big) v. \tag{7}$$

For the maximization over v to remain bounded, we require $\lambda \in \Delta_k$, where

$$\Delta_K := \left\{ \lambda \in \mathbb{R}_+^K : \sum_{k=1}^{K} \lambda_k = 1 \right\} \tag{8}$$

denotes the probability simplex.

Substituting this condition, we obtain the following saddle-point formulation:

$$\min_{\lambda \in \Delta_K} \max_{\pi} \sum_{k=1}^{K} \lambda_k J_k(\pi). \tag{9}$$

Here, the dual variable λ defines a distribution over environments, and the policy π maximizes a weighted sum of returns. This formulation enables the dual variable to adaptively emphasize environments that limit worst-case performance, as detailed in Section III-B.

Remark 1: In general, strong duality does not hold due to the non-convexity of policy optimization in reinforcement learning. Therefore, (9) can be interpreted as a convex relaxation of the original robust objective, and the gap between the two objectives corresponds to the duality gap induced by the non-convexity of policy optimization. Nevertheless, weak duality always holds:

$$\max_{\pi} \min_{k \in [K]} J_k(\pi) \leq \min_{\lambda \in \Delta_K} \max_{\pi} \sum_{k=1}^{K} \lambda_k J_k(\pi).$$

Strong duality can be recovered in structured settings, such as tabular discounted problems formulated over occupancy measures, which yield linear programs. In such settings, the saddle-point formulation (9) is not merely a tractable surrogate but an exact equivalent of the minimax objective (4).

B. Primal–Dual Updates

Based on the saddle-point formulation in (9), we consider an alternating primal–dual optimization procedure.

Primal update (policy update). Given the current dual weights $\lambda^{(m)}$, the policy is updated to maximize the weighted objective:

$$\pi^{(m+1)} = \arg\max_{\pi} \sum_{k=1}^{K} \lambda_k^{(m)} J_k(\pi). \tag{10}$$

In practice, this maximization is approximated via policy gradient updates, whose specific form is detailed in the next section.

Dual update (environment reweighting). After evaluating the environment-wise returns $J_k(\pi^{(m+1)})$, the dual variables are updated via projected gradient descent on the negative return vector, which assigns larger weights to environments with lower returns:

$$\lambda^{(m+1)} = \Pi_{\Delta_K} \left(\lambda^{(m)} - \eta_\lambda \begin{bmatrix} J_1(\pi^{(m+1)}) \\ \vdots \\ J_K(\pi^{(m+1)}) \end{bmatrix} \right), \tag{11}$$

where Π_{Δ_K} denotes projection onto the simplex.

While (9) resembles a standard saddle-point problem, it admits a distinct interpretation in robust reinforcement learning. The worst-case environment is generally *policy-dependent*, meaning that the index attaining $\min_k J_k(\pi)$ may change as the policy evolves. As a result, the problem cannot be reduced to optimizing a fixed environment, but instead requires improving performance on environments that currently limit the objective.

From this viewpoint, the dual variable λ acts as an adaptive reweighting mechanism: it assigns larger weights to environments with lower returns, steering the policy toward improving worst-case performance. The policy π is in turn optimized under the dynamically reweighted objective, focusing on the environments currently emphasized by λ. This closed-loop interaction constitutes the *adaptive minimax optimization* process underlying AMPO.

IV. Policy Gradient for Adaptive Minimax FRL

We develop a policy gradient method to solve the primal update (10) within the proposed primal–dual framework. The primal update requires maximizing a weighted sum of returns across heterogeneous environments:

$$\max_{\pi} \; J_\lambda(\pi) := \sum_{k=1}^{K} \lambda_k J_k(\pi), \tag{12}$$

where $\lambda \in \Delta_K$ is fixed.

Value-based methods such as Q-learning are not directly applicable here: the weighted objective (12) does not correspond to a single MDP, and the worst-case objective further induces a policy-dependent, non-stationary landscape that precludes a well-defined Bellman update. Policy gradient methods, by contrast, optimize directly over trajectories without requiring a unified Bellman structure, making them well-suited for heterogeneous FRL settings.

The weighted objective (12) admits a natural stochastic interpretation: at the beginning of each episode, an environment index k is sampled according to λ, and a trajectory τ is collected by executing π under dynamics P_k. Under this sampling process, we define

$$J_\lambda(\pi) = \mathbb{E}_{k\sim\lambda,\ \tau\sim(\pi,P_k)}\left[R(\tau)\right], \tag{13}$$

where $R(\tau) := \sum_{t=0}^{T-1} \gamma^t r_t$.

Theorem 1 (Extended Policy Gradient Theorem): Let $\lambda \in \Delta_K$ be fixed, and let $\pi_\theta(a|s)$ be a continuously differentiable stochastic policy with full support. Under standard regularity conditions that justify exchanging the gradient and expectation via the Dominated Convergence Theorem, the gradient of $J_\lambda(\pi_\theta)$ is given by

$$\nabla_\theta J_\lambda(\pi_\theta) = \sum_{k=1}^{K} \lambda_k \mathbb{E}_{\tau\sim(\pi_\theta,P_k)}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)\, G_t^{(k)}\right], \tag{14}$$

where $G_t^{(k)} := \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$ denotes the return-to-go at time t under environment k.

Theorem 1 shows that the gradient decomposes into a weighted sum of environment-wise policy gradients. This decomposition aligns naturally with the FRL paradigm: each client k computes a local gradient estimate, and the server aggregates them with weights $\lambda^{(m)}$ determined by the dual update. Since the worst-performing environment is policy-dependent and may shift as π_θ evolves, the dual variable adaptively assigns larger weights to environments with lower returns. This induces an *adaptive reweighting* of the policy gradient in (14), enabling implicit online tracking of the worst-case environment and promoting robust performance across heterogeneous environments.

V. AMPO: Adaptive Minimax Policy Optimization for Robust FRL

Building on the primal–dual formulation in (9), AMPO alternates between policy optimization and environment reweighting under a client–server architecture. At each iteration, clients compute local policy gradients and return estimates, and the server aggregates these signals to update the global policy and dual weights. This closed-loop interaction enables adaptive tracking of the policy-dependent worst-case environment, focusing learning on performance-limiting regimes.

A. The AMPO Algorithm

The proposed AMPO algorithm proceeds by alternating between local updates at the clients and global updates at the server, following a primal–dual optimization procedure.

Local Update (Clients). At iteration m, the server broadcasts the current global policy parameter $\theta^{(m)}$ to all clients. Each client k interacts with its local environment P_k using the policy $\pi_{\theta^{(m)}}$ to collect trajectories. Based on these trajectories, each client computes a sample-based estimate of the local policy gradient:

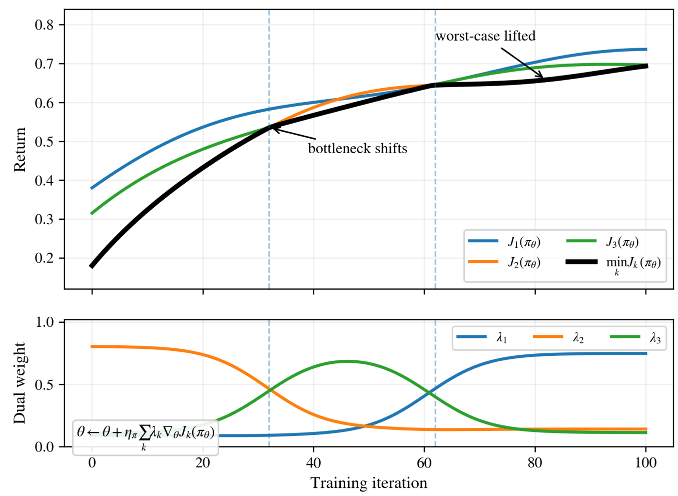

Fig. 1: Illustration of adaptive dual reweighting in the proposed framework. During training, the dual variables $\{\lambda_k\}$ emphasize the current bottleneck environment, guiding policy updates that improve its performance. As the worst-performing environment changes, the minimum return $\min_k J_k(\pi_\theta)$ increases, resulting in a balanced and robust policy across environments.

$$g_k^{(m)} \approx \mathbb{E}_{\tau\sim(\pi_{\theta^{(m)}},P_k)}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_{\theta^{(m)}}(a_t|s_t)\, \hat{G}_t^{(k)}\right], \tag{15}$$

where $\hat{G}_t^{(k)}$ denotes a sample estimate of the return-to-go $G_t^{(k)}$ or a baseline-corrected advantage under environment P_k, obtained from the collected trajectory. Each client also estimates its local return

$$J_k^{(m)} := \widehat{J}_k(\pi_{\theta^{(m)}}), \tag{16}$$

via Monte Carlo rollouts, and transmits $\left(g_k^{(m)}, J_k^{(m)}\right)$ to the server.

Global Update (Server). Upon receiving $\{g_k^{(m)}, J_k^{(m)}\}_{k=1}^{K}$, the server performs the following updates:

- Policy update:

$$g^{(m)} = \sum_{k=1}^{K} \lambda_k^{(m)} g_k^{(m)}, \quad \theta^{(m+1)} = \theta^{(m)} + \eta_\pi\, g^{(m)}. \tag{17}$$

This corresponds to gradient ascent on the dynamically reweighted objective.

- Dual update:

$$\lambda^{(m+1)} = \Pi_{\Delta_K}\left(\lambda^{(m)} - \eta_\lambda \begin{bmatrix} J_1^{(m)} \\ \vdots \\ J_K^{(m)} \end{bmatrix}\right). \tag{18}$$

This update increases the weights of environments with lower returns, encouraging improvement in the worst-performing environments.

Note that λ is not a fixed aggregation coefficient or a manually tuned importance weight, but an optimization variable derived from the minimax formulation (9). Therefore, the weighted gradient aggregation in AMPO should be viewed as the primal update of a saddle-point problem, rather than as a heuristic weighted averaging scheme.

B. Convergence and Regret Analysis

We analyze the convergence behavior of AMPO from the perspective of stochastic saddle-point optimization. For notational convenience, we write the objective in (9) as

$$F(\theta, \lambda) := \sum_{k=1}^{K} \lambda_k J_k(\pi_\theta), \qquad \lambda \in \Delta_K, \tag{19}$$

where π_θ denotes the policy parameterized by θ. The algorithm performs stochastic gradient ascent on θ and projected gradient descent on λ. To facilitate the analysis, we adopt the following standard assumptions. Let $\mathcal{H}_m$ denote the filtration containing all randomness up to the beginning of iteration m.

Assumption 1 (Smoothness): Each $J_k(\pi_\theta)$ is L-smooth in θ.

Assumption 2 (Unbiased gradient with bounded variance): The policy gradient estimator $g^{(m)}$ is conditionally unbiased with bounded variance:

$$\mathbb{E}[g^{(m)} \mid \mathcal{H}_m] = \nabla_\theta F(\theta^{(m)}, \lambda^{(m)}),$$
$$\mathbb{E}\Big[\|g^{(m)} - \nabla_\theta F(\theta^{(m)}, \lambda^{(m)})\|^2 \mid \mathcal{H}_m\Big] \le \sigma^2.$$

Assumption 3 (Unbiased return estimator): The return estimator $\hat{J}_k^{(m)}$ is conditionally unbiased with bounded second moment:

$$\mathbb{E}[\hat{J}_k^{(m)} \mid \mathcal{H}_m] = J_k(\pi_{\theta^{(m)}}),$$
$$\mathbb{E}\left[\left|\hat{J}_k^{(m)} - J_k(\pi_{\theta^{(m)}})\right|^2 \mid \mathcal{H}_m\right] \le \sigma_J^2.$$

Assumption 4 (Bounded domain): The iterates $\{(\theta^{(m)}, \lambda^{(m)})\}$ remain in a bounded domain.

We now characterize the convergence behavior of the proposed primal–dual updates under Assumptions 1–4.

Theorem 2 (Primal Convergence and Dual No-Regret): Under Assumptions 1–4, with step sizes $\eta_\pi = \mathcal{O}(M^{-1/3})$ and $\eta_\lambda = \mathcal{O}(M^{-2/3})$, the iterates generated by AMPO satisfy

$$\frac{1}{M}\sum_{m=0}^{M-1} \mathbb{E}\left[\left\|\nabla_\theta F(\theta^{(m)}, \lambda^{(m)})\right\|^2\right] = \mathcal{O}\Big(M^{-1/3}\Big),$$
$$\sum_{m=0}^{M-1} F(\theta^{(m)}, \lambda^{(m)}) - \min_{\lambda \in \Delta_K} \sum_{m=0}^{M-1} F(\theta^{(m)}, \lambda) = \mathcal{O}\Big(M^{2/3}\Big).$$

The dual regret rate $\mathcal{O}(M^{2/3})$, slower than the classical $\mathcal{O}(\sqrt{M})$, is a consequence of the two-timescale step size choice $\eta_\pi = \mathcal{O}(M^{-1/3})$ and $\eta_\lambda = \mathcal{O}(M^{-2/3})$, which is necessitated by the primal–dual coupling. Specifically, achieving $\mathcal{O}(M^{-1/3})$ primal convergence requires $\eta_\lambda/\eta_\pi \to 0$, which in turn inflates the dual regret. This reflects an inherent trade-off between primal convergence speed and dual tracking accuracy. Notably, when the environment weights are fixed, the problem reduces to standard policy optimization with a static objective, for which the classical $\mathcal{O}(1/\sqrt{M})$ convergence rate can be recovered. The slower rate in our result thus reflects the additional challenge of adaptive, policy-dependent reweighting.

The first result shows that the policy converges to a first-order stationary point of a *dynamically reweighted objective*. Although the objective is non-stationary due to the evolving dual weights, convergence to a first-order stationary point of the time-varying objective $F(\theta, \lambda^{(m)})$ ensures that the time-averaged gradient norm vanishes asymptotically, implying that the learning dynamics remain stable under adaptive reweighting. We note that this stationary point is established with respect to $F(\theta, \lambda^{(m)})$ rather than the original minimax objective $\max_\pi \min_k J_k(\pi)$; closing this gap in general non-convex settings remains an open problem. In structured settings where strong duality holds (Remark 1), however, the stationary point of F coincides with that of the minimax objective.

The second result establishes that the dual updates achieve sublinear regret over the simplex Δ_K, implying that the adaptive weighting $\{\lambda^{(m)}\}$ performs nearly as well as the best fixed weighting in hindsight. Consequently, the algorithm avoids persistently focusing on suboptimal environments and instead emphasizes those that limit performance. Taken together, these results provide a principled justification for AMPO: the policy steadily improves on the hardest environments while the dual weighting reliably tracks them—all established in an ergodic sense standard in stochastic saddle-point optimization. While this does not imply last-iterate optimality, it ensures that the time-averaged behavior approaches a robust solution in terms of worst-case performance.

Remark 2 (Practical choice of learning rates): The rates in Theorem 2 suggest a practical two-timescale update rule for AMPO. Since $\eta_\lambda = \mathcal{O}(M^{-2/3})$ decays faster than $\eta_\pi = \mathcal{O}(M^{-1/3})$, the theoretical scaling requires $\eta_\lambda/\eta_\pi = \mathcal{O}(M^{-1/3}) \to 0$, implying that the dual variable should be updated strictly more conservatively than the policy parameter. In practice, this motivates choosing $\eta_\lambda \ll \eta_\pi$. The dual step size η_λ governs a fundamental trade-off. A large η_λ causes the weights λ to react too aggressively to noisy return estimates: the worst-case environment index may oscillate across iterations, destabilizing the policy update direction and preventing reliable tracking of performance-limiting environments. Conversely, an overly small η_λ renders λ nearly static, reducing AMPO to uniform aggregation and forfeiting the robustness benefits of adaptive reweighting. Thus, the theoretical step-size scaling supports using a moderately smaller dual learning rate, and we recommend treating the ratio η_λ/η_π as the primary tuning parameter rather than η_λ and η_π independently.

C. Robustness–Scalability Trade-off

We analyze the impact of adaptive dual weighting on the scalability of federated policy optimization. In standard FRL methods that optimize an average-return objective, local gradients are uniformly averaged across K clients, achieving variance reduction proportional to $1/K$ and enabling linear speedup in K. In contrast, AMPO aggregates gradients using adaptive weights:

$$\nabla_\theta F(\theta, \lambda) = \sum_{k=1}^{K} \lambda_k \nabla_\theta J_k(\pi_\theta), \tag{20}$$

where $\lambda \in \Delta_K$ evolves to emphasize environments with lower returns.

Proposition 1 (Effective Parallelism under Adaptive Weighting): Under the conditional independence assumption on local gradient estimators $\{g_k\}_{k=1}^K$ given $\mathcal{H}_m$, and assuming each estimator is unbiased with bounded per-client variance σ_u^2, the variance of the aggregated gradient satisfies

$$\mathbb{E}\left[\left\| \sum_{k=1}^{K} \lambda_k (g_k - \nabla_\theta J_k) \right\|^2 \,\middle|\, \mathcal{H}_m \right] \;\le\; \frac{\sigma_u^2}{K_{\text{eff}}}, \tag{21}$$

where the *effective number of environments* is defined as

$$K_{\text{eff}} := \frac{1}{\sum_{k=1}^{K} \lambda_k^2} \;\in\; [1,\, K]. \tag{22}$$

The quantity K_{eff} characterizes the effective degree of variance reduction in the aggregated gradient. When λ is uniform, i.e., $\lambda_k = 1/K$, we recover $K_{\text{eff}} = K$, corresponding to classical linear speedup. As λ concentrates on a subset of environments, K_{eff} decreases toward 1, reducing effective variance reduction.

We now make the connection to Theorem 2. The aggregated gradient is given by $g^{(m)} = \sum_{k=1}^{K} \lambda_k^{(m)} g_k^{(m)}$, where each local estimator $g_k^{(m)}$ has per-client variance bounded by σ_u^2. By Proposition 1, the variance of the aggregated gradient satisfies

$$\mathbb{E}\big[\|g^{(m)} - \nabla_\theta F_m\|^2 \mid \mathcal{H}_m\big] \;\le\; \frac{\sigma_u^2}{K_{\text{eff}}^{(m)}} \;\le\; \sigma_u^2 \;=:\; \sigma^2, \tag{23}$$

where the second inequality follows from $K_{\text{eff}}^{(m)} \ge 1$ for all m. This confirms that Assumption 2 is satisfied with $\sigma^2 = \sigma_u^2$, and the convergence guarantee of Theorem 2 holds. Moreover, the tighter per-iteration bound $\sigma_u^2 / K_{\text{eff}}^{(m)}$ reveals that the effective noise level is strictly smaller than σ^2 whenever $K_{\text{eff}}^{(m)} > 1$, i.e., when the dual weights are not fully concentrated on a single environment.

Since $\sigma^2 = \sigma_u^2$ by (23), substituting the tighter per-iteration bound $\sigma_u^2 / K_{\text{eff}}^{(m)}$ from Proposition 1 into the proof of Theorem 2, the noise term $L\eta_\pi \sigma^2$ in (38) refines to $L\eta_\pi \sigma_u^2 / \bar{K}_{\text{eff}}$, yielding:

$$\frac{1}{M} \sum_{m=0}^{M-1} \mathbb{E}\|\nabla_\theta F_m\|^2 \;\le\; \frac{2C_2}{M\eta_\pi} + \frac{L\eta_\pi \sigma_u^2}{\bar{K}_{\text{eff}}} + \frac{2J_{\max} C_\lambda \, \eta_\lambda}{\eta_\pi}, \tag{24}$$

where

$$\bar{K}_{\text{eff}} := \left(\frac{1}{M} \sum_{m=0}^{M-1} \frac{1}{K_{\text{eff}}^{(m)}} \right)^{-1} \tag{25}$$

is the harmonic mean of $\{K_{\text{eff}}^{(m)}\}$ over iterations, satisfying $1 \le \bar{K}_{\text{eff}} \le K$. Choosing $\eta_\pi = \mathcal{O}(M^{-1/3})$ and $\eta_\lambda = \mathcal{O}(M^{-2/3})$, the dominant term in (24) yields an effective primal convergence rate of $\mathcal{O}(\bar{K}_{\text{eff}}^{-1} M^{-1/3})$. In the uniform case $\lambda_k = 1/K$, we have $\bar{K}_{\text{eff}} = K$ and the standard linear speedup $\mathcal{O}(K^{-1} M^{-1/3})$ is recovered. Under adaptive reweighting, $\bar{K}_{\text{eff}} < K$, and the speedup is only partial, with the ratio $K/\bar{K}_{\text{eff}} \ge 1$ quantifying the robustness cost relative to uniform aggregation.

This analysis reveals a fundamental trade-off: adaptive reweighting improves robustness by emphasizing challenging environments, but reduces effective parallelism by concentrating gradient mass on a subset of clients. Consequently, AMPO achieves a *partial speedup* governed by $\bar{K}_{\text{eff}}$ rather than full linear speedup in K. This trade-off is intrinsic to adaptive minimax optimization: prioritizing difficult environments improves robustness but necessarily increases gradient variance, and the ratio $K/\bar{K}_{\text{eff}} \ge 1$ provides a quantitative characterization of the robustness–scalability trade-off in FRL.

VI. PPO-Based Implementation of AMPO

We present a practical instantiation of the proposed AMPO framework using Proximal Policy Optimization (PPO) within an actor–critic architecture. While we adopt PPO as a stable and widely validated backbone, the AMPO framework is algorithm-agnostic and can be combined with other policy optimization methods, including TRPO, A2C/A3C, and SAC, with minimal modification.

We consider a shared stochastic policy $\pi_\theta(a \mid s)$ parameterized by θ, which is maintained by the server and broadcast to all clients. Each client k interacts with its local environment P_k and maintains an environment-specific critic V_{ψ_k} for value estimation. This design is well suited to FRL, where clients may have heterogeneous transition dynamics and reward landscapes. Unlike multi-agent RL methods such as MAPPO, which typically assume agents interacting within a shared environment and rely on a centralized critic, our setting consists of multiple heterogeneous environments. Thus, local critics naturally capture client-specific value functions while sharing a common robust policy.

Local Update (Clients). At iteration m, each client k receives the global policy parameter $\theta^{(m)}$ from the server and collects trajectories by interacting with its local environment P_k. Each client updates its local critic V_{ψ_k} via temporal-difference (TD) learning. Based on the collected trajectories, the advantage is estimated using generalized advantage estimation (GAE):

$$A_t^{(k)} = \sum_{l=0}^{T-1-t} (\gamma \lambda_{\text{GAE}})^l \delta_{t+l}^{(k)}, \tag{26}$$

$$\delta_t^{(k)} = r_t + \gamma V_{\psi_k}(s_{t+1}) - V_{\psi_k}(s_t), \tag{27}$$

where $\lambda_{\text{GAE}} \in [0,1]$ controls the bias–variance trade-off. Using the estimated advantages, each client forms the PPO surrogate objective:

$$\mathcal{L}_k^{\text{PPO}}(\theta) = \mathbb{E}\left[\min\Big(r_t(\theta) A_t^{(k)}, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\, A_t^{(k)}\Big)\right], \tag{28}$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta^{(m)}}(a_t \mid s_t)}. \tag{29}$$

Each client performs multiple PPO updates based on mini-batches of collected trajectories and computes a stochastic policy gradient estimate

$$g_k^{(m)} = \nabla_\theta \mathcal{L}_k^{\mathrm{PPO}}(\theta)\big|_{\theta=\theta^{(m)}}. \tag{30}$$

In addition, each client estimates the local return

$$J_k^{(m)} := \widehat{J}_k(\pi_{\theta^{(m)}}), \tag{31}$$

via Monte Carlo rollouts, and transmits $\left(g_k^{(m)}, J_k^{(m)}\right)$ to the server.

Remark 3 (PPO surrogate and Assumption 2): The clipped PPO surrogate introduces a bounded bias relative to the true policy gradient; in practice, this bias is controlled by the clipping parameter ϵ and diminishes as the policy update magnitude decreases, making Assumption 2 approximately satisfied throughout training.

Global Update (Server). Upon receiving $\{g_k^{(m)}, J_k^{(m)}\}_{k=1}^K$ from all clients, the server performs the policy and dual updates.

i) Policy update. The server aggregates local gradients using the current dual weights:

$$g^{(m)} = \sum_{k=1}^{K} \lambda_k^{(m)} g_k^{(m)}, \tag{32}$$

$$\theta^{(m+1)} = \theta^{(m)} + \eta_\pi g^{(m)}. \tag{33}$$

This corresponds to gradient ascent on a dynamically reweighted objective, where environments with larger dual weights contribute more to the update.

ii) Dual update. The server updates the dual variable based on the observed returns:

$$\lambda^{(m+1)} = \Pi_{\Delta_K}\left(\lambda^{(m)} - \eta_\lambda \begin{bmatrix} J_1^{(m)} \\ \vdots \\ J_K^{(m)} \end{bmatrix}\right), \tag{34}$$

where $\eta_\lambda > 0$ is the dual step size and Π_{Δ_K} denotes projection onto the probability simplex. This update steers the shared policy toward improving worst-case performance by assigning larger weights to environments with lower returns, realizing the adaptive minimax optimization behavior of AMPO.

VII. Experiments

A. Experimental Setup

B. Worst-Case Performance Analysis

C. Overall Performance under Heterogeneous Environments

D. Dual Variable Dynamics

E. Ablation Study

VIII. Conclusion

Appendix A
Proof of Theorem 1

Recall that the weighted objective is defined as

$$J_\lambda(\pi_\theta) = \sum_{k=1}^{K} \lambda_k J_k(\pi_\theta),$$

Algorithm 1 AMPO with PPO Backbone

Require: Initial actor parameter $\theta^{(0)}$, dual variable $\lambda^{(0)} \in \Delta_K$, number of rounds M

for $m = 0, 1, \ldots, M-1$ **do**
  Server broadcast: send $\theta^{(m)}$ to all clients
  for each client k **in parallel do**
    Collect trajectories in environment P_k using $\pi_{\theta^{(m)}}$
    Update local critic V_{ψ_k} via TD learning
    Compute advantages $A_t^{(k)}$ via GAE (26)–(27)
    Perform multiple PPO updates; obtain a stochastic gradient estimate $g_k^{(m)}$ via (30)
    Estimate return $J_k^{(m)}$ via Monte Carlo rollouts
    Send $\left(g_k^{(m)}, J_k^{(m)}\right)$ to the server
  end for
  Server update:
  Perform policy update via (32) and (33)
  Perform dual update via (34)
  Broadcast updated policy $\theta^{(m+1)}$ to all clients
end for

where

$$J_k(\pi_\theta) = \mathbb{E}_{\tau\sim(\pi_\theta, P_k)}\left[R(\tau)\right].$$

Since $\lambda \in \Delta_K$ is fixed and independent of θ, we can differentiate term-wise. Under standard regularity conditions—specifically, bounded rewards, a continuously differentiable policy with full support, and $\mathbb{E}\left[\left\|\nabla_\theta \log \pi_\theta(a_t|s_t)\, G_t^{(k)}\right\|\right] < \infty$—which justify exchanging the gradient and expectation via the Dominated Convergence Theorem, we obtain

$$\nabla_\theta J_\lambda(\pi_\theta) = \sum_{k=1}^{K} \lambda_k \nabla_\theta J_k(\pi_\theta).$$

For each environment k, applying the standard policy gradient theorem yields

$$\nabla_\theta J_k(\pi_\theta) = \mathbb{E}_{\tau\sim(\pi_\theta, P_k)}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)\, G_t^{(k)}\right],$$

where $G_t^{(k)} := \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$ denotes the return-to-go under environment k. Substituting this expression into the weighted gradient yields

$$\nabla_\theta J_\lambda(\pi_\theta) = \sum_{k=1}^{K} \lambda_k \mathbb{E}_{\tau\sim(\pi_\theta, P_k)}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)\, G_t^{(k)}\right],$$

which is exactly (14). This completes the proof.

Appendix B
Proof of Theorem 2

We prove the primal convergence and dual no-regret results separately.

A. Primal Convergence

Let $\mathcal{H}_m$ denote the filtration containing all randomness up to the beginning of iteration m, and define

$$F_m := F(\theta^{(m)}, \lambda^{(m)}).$$

Step 1: One-step descent via L-smoothness. Since $F(\theta,\lambda)$ is L-smooth in θ (Assumption 1), the policy update $\theta^{(m+1)} = \theta^{(m)} + \eta_\pi g^{(m)}$ gives

$$F(\theta^{(m+1)},\lambda^{(m)}) \;\ge\; F_m + \eta_\pi\langle\nabla_\theta F_m,\, g^{(m)}\rangle - \frac{L\eta_\pi^2}{2}\|g^{(m)}\|^2.$$

Taking the conditional expectation over $\mathcal{H}_m$ and using Assumption 2($\mathbb{E}[g^{(m)} \mid \mathcal{H}_m] = \nabla_\theta F_m$ and $\mathbb{E}[\|g^{(m)}\|^2 \mid \mathcal{H}_m] \le \|\nabla_\theta F_m\|^2 + \sigma^2$), we can get:

$$\begin{aligned}
&\mathbb{E}\big[F(\theta^{(m+1)},\lambda^{(m)})\big]\\
&\ge\; \mathbb{E}[F_m] + \eta_\pi\mathbb{E}\|\nabla_\theta F_m\|^2 - \frac{L\eta_\pi^2}{2}\big(\mathbb{E}\|\nabla_\theta F_m\|^2 + \sigma^2\big)\\
&\ge\; \mathbb{E}[F_m] + \frac{\eta_\pi}{2}\mathbb{E}\|\nabla_\theta F_m\|^2 - \frac{L\eta_\pi^2}{2}\sigma^2, \qquad (35)
\end{aligned}$$

where the last inequality uses $\eta_\pi \le 1/L$ so that $\eta_\pi - L\eta_\pi^2/2 \ge \eta_\pi/2$.

Step 2: Accounting for the dual update. Since $F(\theta,\lambda) = \sum_{k=1}^K \lambda_k J_k(\pi_\theta)$ is linear in λ and the returns are uniformly bounded by $J_{\max}$, we have

$$\begin{aligned}
&\big|F(\theta^{(m+1)},\lambda^{(m+1)}) - F(\theta^{(m+1)},\lambda^{(m)})\big|\\
&\quad\le\; J_{\max}\|\lambda^{(m+1)} - \lambda^{(m)}\|_1.
\end{aligned}$$

Since the projection Π_{Δ_K} is non-expansive and $\|\hat g_\lambda^{(m)}\|_2 \le \sqrt{K}J_{\max}$ (from the uniform boundness of returns, since $|J_k^{(m)}| \le J_{\max}$ for all k), the dual step satisfies

$$\begin{aligned}
\|\lambda^{(m+1)} - \lambda^{(m)}\|_1 \;&\le\; \sqrt{K}\|\lambda^{(m+1)} - \lambda^{(m)}\|_2\\
&\le\; \sqrt{K}\cdot\eta_\lambda\|\hat g_\lambda^{(m)}\|_2 \;\le\; KJ_{\max}\,\eta_\lambda\\
&=:\; C_\lambda\,\eta_\lambda.
\end{aligned}$$

Therefore, we obtain:

$$\mathbb{E}\big[F(\theta^{(m+1)},\lambda^{(m+1)})\big] \;\ge\; \mathbb{E}\big[F(\theta^{(m+1)},\lambda^{(m)})\big] - J_{\max}C_\lambda\,\eta_\lambda. \qquad (36)$$

Step 3: Combining Steps 1 and 2. Adding (35) and (36), we obtain

$$\begin{aligned}
\mathbb{E}[F(\theta^{(m+1)},\lambda^{(m+1)})] \;\ge\; &\mathbb{E}[F_m] + \frac{\eta_\pi}{2}\mathbb{E}\|\nabla_\theta F_m\|^2\\
&- \frac{L\eta_\pi^2}{2}\sigma^2 - J_{\max}C_\lambda\,\eta_\lambda. \qquad (37)
\end{aligned}$$

Step 4: Telescoping and averaging. Rearranging (37) and summing over $m = 0,\dots,M-1$, we obtain the upper bound:

$$\begin{aligned}
\frac{\eta_\pi}{2}\sum_{m=0}^{M-1}\mathbb{E}\|\nabla_\theta F_m\|^2 \;\le\; &\sum_{m=0}^{M-1}\mathbb{E}\big[F(\theta^{(m+1)},\lambda^{(m+1)}) - F_m\big]\\
&+ \frac{L\eta_\pi^2\sigma^2 M}{2} + J_{\max}C_\lambda\eta_\lambda M.
\end{aligned}$$

The sum on the right-hand side is a telescoping sum:

$$\begin{aligned}
&\sum_{m=0}^{M-1}\mathbb{E}\big[F(\theta^{(m+1)},\lambda^{(m+1)}) - F(\theta^{(m)},\lambda^{(m)})\big]\\
&= \mathbb{E}[F(\theta^{(M)},\lambda^{(M)})] - F(\theta^{(0)},\lambda^{(0)}) \;\le\; C_2,
\end{aligned}$$

where $C_2 > 0$ is a finite constant that exists because the iterates remain in a bounded domain (Assumption 4) and F is continuous. Dividing both sides by $\eta_\pi M/2$, we have:

$$\begin{aligned}
\frac{1}{M}\sum_{m=0}^{M-1}\mathbb{E}\|\nabla_\theta F_m\|^2 \;&\le\; \frac{2C_2}{M\eta_\pi} + L\eta_\pi\sigma^2 + \frac{2J_{\max}C_\lambda\,\eta_\lambda}{\eta_\pi}\\
&=\; \mathcal{O}\bigg(\frac{1}{M\eta_\pi} + \eta_\pi + \frac{\eta_\lambda}{\eta_\pi}\bigg). \qquad (38)
\end{aligned}$$

Choosing $\eta_\pi = \mathcal{O}(M^{-1/3})$ and $\eta_\lambda = \mathcal{O}(M^{-2/3})$ balances the three terms in (38):

$$\frac{1}{M\eta_\pi} = \mathcal{O}(M^{-2/3}), \quad \eta_\pi = \mathcal{O}(M^{-1/3}), \quad \frac{\eta_\lambda}{\eta_\pi} = \mathcal{O}(M^{-1/3}),$$

which gives

$$\frac{1}{M}\sum_{m=0}^{M-1}\mathbb{E}\|\nabla_\theta F_m\|^2 \;=\; \mathcal{O}\Big(M^{-1/3}\Big).$$

This establishes the primal convergence claim.

B. Dual No-Regret

For each round m, define the per-round loss as

$$f_m(\lambda) := F(\theta^{(m)},\lambda) = \sum_{k=1}^K \lambda_k J_k(\pi_{\theta^{(m)}}).$$

Conditioned on $\mathcal{H}_m$, the policy parameter $\theta^{(m)}$ is fixed, so $f_m(\lambda)$ is a *linear* (hence convex) function of λ over Δ_K.

Stochastic gradient properties. The dual gradient estimator is $\hat g_\lambda^{(m)} = [J_1^{(m)},\dots,J_K^{(m)}]^\top$. By Assumption 3 it satisfies

$$\mathbb{E}\big[\hat g_\lambda^{(m)} \mid \mathcal{H}_m\big] = \nabla_\lambda f_m(\lambda^{(m)}), \qquad \mathbb{E}\big[\|\hat g_\lambda^{(m)}\|^2 \mid \mathcal{H}_m\big] \;\le\; G^2,$$

for some constant $G > 0$ (which follows from the uniform boundedness of returns and $\gamma < 1$).

Online convex optimization regret bound. The dual update

$$\lambda^{(m+1)} = \Pi_{\Delta_K}\Big(\lambda^{(m)} - \eta_\lambda\hat g_\lambda^{(m)}\Big)$$

is a step of projected stochastic gradient descent on the sequence of convex functions $\{f_m\}$. By the standard analysis of online projected gradient descent [8], [9], for any fixed comparator $\lambda \in \Delta_K$:

$$\begin{aligned}
&\mathbb{E}\left[\sum_{m=0}^{M-1}\big(f_m(\lambda^{(m)}) - f_m(\lambda)\big)\right]\\
&\quad\le\; \frac{\|\lambda - \lambda^{(0)}\|^2}{2\eta_\lambda} + \frac{\eta_\lambda}{2}\sum_{m=0}^{M-1}\mathbb{E}\|\hat g_\lambda^{(m)}\|^2\\
&\quad\le\; \frac{D^2}{2\eta_\lambda} + \frac{\eta_\lambda G^2 M}{2},
\end{aligned}$$

where $D := \max_{\lambda,\lambda'\in\Delta_K}\|\lambda - \lambda'\|$ is the diameter of Δ_K.

Choosing $\eta_\lambda = \mathcal{O}(M^{-2/3})$, we have:

$$\frac{D^2}{2\eta_\lambda} = \mathcal{O}(M^{2/3}), \qquad \frac{\eta_\lambda G^2 M}{2} = \mathcal{O}(M^{1/3}).$$

Hence the dominant term gives

$$\mathbb{E}\left[\sum_{m=0}^{M-1}\big(f_m(\lambda^{(m)})-f_m(\lambda)\big)\right]=\mathcal{O}\Big(M^{2/3}\Big).$$

Since this holds for every fixed $\lambda \in \Delta_K$, it holds in particular for the best fixed comparator in hindsight. Substituting $f_m(\lambda)=F(\theta^{(m)},\lambda)$:

$$\mathbb{E}\left[\sum_{m=0}^{M-1}F(\theta^{(m)},\lambda^{(m)})-\min_{\lambda\in\Delta_K}\sum_{m=0}^{M-1}F(\theta^{(m)},\lambda)\right]=\mathcal{O}\Big(M^{2/3}\Big).$$

This establishes the dual no-regret claim and completes the proof.

Appendix C
Proof of Proposition 1

Let $\xi_k := g_k - \nabla_\theta J_k(\pi_\theta)$. Under the per-client variance assumption, we have

$$\mathbb{E}[\xi_k \mid \mathcal{H}_m]=0, \qquad \mathbb{E}\big[\|\xi_k\|^2 \mid \mathcal{H}_m\big]\le \sigma_u^2.$$

Expanding the squared norm yields

$$\begin{aligned}\mathbb{E}\left[\left\|\sum_{k=1}^{K}\lambda_k\xi_k\right\|^2 \,\middle|\, \mathcal{H}_m\right]&=\mathbb{E}\left[\left\langle\sum_{i=1}^{K}\lambda_i\xi_i,\sum_{j=1}^{K}\lambda_j\xi_j\right\rangle \,\middle|\, \mathcal{H}_m\right]\\ &=\sum_{k=1}^{K}\lambda_k^2\mathbb{E}\big[\|\xi_k\|^2\mid\mathcal{H}_m\big]+\sum_{i\neq j}\lambda_i\lambda_j\mathbb{E}\big[\langle\xi_i,\xi_j\rangle\mid\mathcal{H}_m\big].\end{aligned} \tag{39}$$

Under the conditional independence assumption on $\{\xi_k\}_{k=1}^K$ given $\mathcal{H}_m$, the cross terms vanish, i.e.,

$$\mathbb{E}[\langle\xi_i,\xi_j\rangle\mid\mathcal{H}_m]=0, \quad i\neq j.$$

Therefore, we have

$$\begin{aligned}\mathbb{E}\left[\left\|\sum_{k=1}^{K}\lambda_k\xi_k\right\|^2 \,\middle|\, \mathcal{H}_m\right]&=\sum_{k=1}^{K}\lambda_k^2\mathbb{E}\big[\|\xi_k\|^2\mid\mathcal{H}_m\big]\\ &\le\sigma_u^2\sum_{k=1}^{K}\lambda_k^2=\frac{\sigma_u^2}{K_{\text{eff}}},\end{aligned} \tag{40}$$

where the last equality follows from the definition $K_{\text{eff}} := 1/\sum_{k=1}^{K}\lambda_k^2$ in (22). This completes the proof.

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